

Start of Assessment

DISCLAIMER: this assessment paper has been prepared to provide a sample of the style and content of questions students may find in the Final Written Assessment. Please note that this is an **abbreviated** paper, containing only one or two questions from each of the 8 main question categories, hence being only out of 26.5 marks.

The actual Final Written Assessment paper will contain more questions, and will typically be marked out of:

- **TNE10006** – 90 to 100 marks
- **TNE60006** – 100 to 110 marks

Q1 Consider the 802.3 Ethernet Protocol.

a) Do collisions occur in a switched network? Why/Why Not?

No, because each device has its collision domain

(3 marks)

(3 marks)

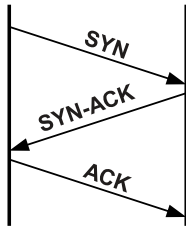
Q2 Consider the IP Protocol

- a) Answer each of the following questions **TRUE** or **FALSE**:
- i. False 57.69.168.31/27 is a valid host IP address (1 mark)
 - ii. True 205.64.87.17 is in the 205.64.87.0/26 subnet (1 mark)
- b) An IP Packet of size 5,730 bytes is sent over a link with a 600 byte MTU
- i. How many IP fragments are sent?
10
(1 mark)
 - ii. Fragment 3 is lost, will the IP layer request a retransmission?
No, because this is not the task of IP layer
(1 mark)
- c) Write the following IPv6 addresses in abbreviated form:
- i. 48a4:00b4:0000:0000:0000:0000:cd00:0a7b
48a4:b4::cd00:a7b
(1 mark)
- d) Consider the host with the IPv6 Address 2001:16d4:b:4:13a1:18ee:ed2b:8f7b/64
- i. What is the Site Address Space ID with prefix?
2001:16d4:b/48
(1 mark)
- (6 marks)**

Q3 Question 3 is a VLSM question worth 15 marks. You should understand the type of question without a sample

Q4 This question concerns Transport Layer Protocols

- a) Consider the TCP Three-Way Handshake depicted in the figure below, the sequence number of the first **SYN** packet is 1,543



- i. How many bytes of data are contained within the first **SYN** Packet?

0B

(1 mark)

- ii. In the **SYN-ACK** response, what is the Acknowledgement number?

1544

(1 mark)

- iii. What is the sequence number in the **SYN-ACK** response?

random or a unit number

(2 marks)

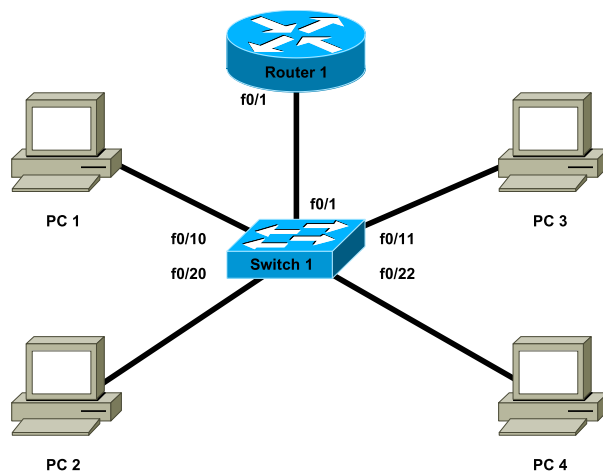
- iv. How many bytes of data may the sender include in the final **ACK** packet?

0B

(1 mark)

(5 marks)

Q5 Consider the following network with associated IP Address, MAC Address and ARP/MAC table information



PC ARP Tables

PC 1

IP	MAC
192.168.10.1	aa:bb:cc:dd:ee:99

PC 2

IP	MAC
Empty	

PC 3

IP	MAC
Empty	

PC 4

IP	MAC
Empty	

Router 1

IP	MAC
192.168.10.6	aa:bb:cc:dd:ff:01

Interface Configuration Details

Device	Interface	VLAN	MAC	IP
Router 1	f0/1.10	10	aa:bb:cc:dd:ee:99	192.168.10.1
	f0/1.20	20	aa:bb:cc:dd:ee:99	192.168.20.1
	f0/1.99	99	aa:bb:cc:dd:ee:99	192.168.99.1
Switch 1	f0/1	Trunk	—	—
	f0/10	10	—	—
	f0/11	10	—	—
	f0/20	20	—	—
	f0/22	20	—	—
	vlan99	99	aa:bb:cc:dd:00:99	192.168.99.5
PC 1	—	—	aa:bb:cc:dd:ff:01	192.168.10.6
PC 2	—	—	aa:bb:cc:dd:ff:02	192.168.20.7
PC 3	—	—	aa:bb:cc:dd:ff:03	192.168.10.8
PC 4	—	—	aa:bb:cc:dd:ff:04	192.168.20.9

Switch 1 MAC Table

MAC	Port
aa:bb:cc:dd:ee:99	f0/1
aa:bb:cc:dd:ff:01	f0/10

- a) When a packet from PC1 to PC4 traverses the trunk link from **Switch 1** to **Router 1**, fill in the following information as seen in the packet headers

	Source	Destination
MAC	aa:bb:cc:dd:ff:01	aa:bb:cc:dd:ee:99
IP	192.168.10.6	192.168.20.9

(2 marks)

- b) Nominate one advantage and one disadvantage of a layered network protocol architecture?

- Advantage: flexibility, scalability, manageability
- Disadvantage: redundancy, increase the complexity

(2 marks)

(4 marks)

Q6 This question relates to the Spanning Tree Protocol

- a) How is it possible to configure Cisco Switches such that a different switch becomes the root bridge for each VLAN?

we can use pvst+

(2 marks)

(2 marks)

Q7 This question refers to aspects of the design of Switched networks

- a) At which layer(s) in a Hierarchical network (*Core, Distribution or Access*) are the following switch features most important (*you may tick more than one layer*)

Switch Feature	Core	Distribution	Access
Power over Ethernet			X

($\frac{1}{2}$ mark)

- b) Describe briefly what the term **Converged Network** means?

Using a properly designed hierarchical network, and implementing QoS policies,...

(1 mark)

($1\frac{1}{2}$ marks)

Q8 This question is about Ethernet Switching and VLANs

- a) Nominate one advantage and one disadvantage to using trunking instead of Access Ports when connecting a Switch to another Switch or Router?

i. Advantage

allow multiple vlans packets can travel via the single physical line

(1 mark)

ii. Disadvantage

Vulnerable to security threats

(1 mark)

- b) Briefly explain how each of the following benefits are realised through the use of VLANs

i. Cost Reduction

reduce the cost of routers,..

(2 marks)

(4 marks)

Q9 Consider a wireless network

a) What purpose does the SSID serve in a Wireless network?

the name of the access points helps to identify the access points

(1 mark)

(1 marks)

End of Assessment

Student Marks – Staff Use Only

Question:	1	2	3	4	5	6	7	8	9	Total
Points:	3	6	0	5	4	2	1½	4	1	26½
Score:										